Strain-Modulated Structural Stability and Dopant-Induced Lattice Distortion in Se-Substituted ZrS_2: A Machine-Learning Potential Study

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Abstract

Customizing the structural stability of layered transition metal dichalcogenides (TMDCs) requires an understanding of how strain engineering and substitutional doping interact. In this work, we use a machine-learned interatomic potential framework to study the strain-dependent energetics and local lattice response of Se-substituted ZrS_2. Both pristine and doped supercells are subjected to a systematic application of biaxial strain between -4 and +4% in order to assess the formation energetics, elastic curvature, and structural relaxation behavior. Elastic stiffness in undoped and Se-substituted systems can be quantitatively compared thanks to the strain–energy response's distinct parabolic pattern. Dopant stability is found to be somewhat sensitive to lattice deformation, according to strain-dependent formation energy analysis. Medium-range elastic relaxation is shown by the bond length distribution and shell-matched distortion analysis, which show a noticeable first-shell expansion (~0.12 Å) surrounding the Se dopant with detectable lattice disruption up to the fourth coordination shell. A coherent atomistic image of dopant-induced lattice alteration in ZrS_2 is provided by the combined mechanical, thermodynamic, and structural analysis, which further emphasizes the usefulness of machine-learning potentials for effective strain-defect investigations.

Keywords: ZrS_2, Transition Metal Dichalcogenides, Strain Engineering, Substitutional Doping, Lattice Distortion, Formation Energy, Machine-Learning Interatomic Potentials

1. Introduction

Two-dimensional layered materials have attracted considerable research attention due to their structural flexibility and the possibility of tuning their physical properties through external perturbations [1-3]. Transition metal dichalcogenides (TMDCs) have attracted considerable attention due to their anisotropic bonding characteristics. It consist of strong in-plane covalent bonds and weak interlayer van der Waals interactions [4]. For flexible and strain-sensitive device topologies, this structural duality offers considerable mechanical flexibility without causing instantaneous structural breakdown [5].

Group-IV TMDCs include zirconium disulfide (ZrS2), which crystallizes in a layered CdI2-type structure [6]. Although ZrS_2 demonstrates a moderate semiconducting band gap along with good chemical stability. It has attracted less research interest than the widely investigated Mo- and W-based dichalcogenides [7, 8]. It is an appropriate model system for examining strain–structure connections at the atomic scale because of its comparatively straightforward layered geometry and mechanical integrity [9]. In practical device applications, these materials are seldom strain-free. In-plane lattice distortions frequently arise from fabrication processes, substrate mismatch, and variations in thermal expansion [10]. Therefore, to comprehend ZrS2 structural robustness, it is crucial to assess its elastic response under controlled biaxial strain [11].

Atomic substitution has become a regulated method to alter local bonding conditions without significantly altering the host lattice, in line with strain engineering [12, 13]. The replacement of sulfur by selenium in chalcogenide systems is also chemically motivated since both elements are similar in terms of valence configurations despite the fact that they vary in terms of their atomic radii. Even small such size differences can cause changes in bond lengths, local lattice relaxation, and the elastic energy distribution in the vicinity of the defect region [14]. More importantly, depending on the rigidity of the lattice and the bonding topology, such a replacement can have more than the first coordination shell perturbed; the elastic perturbations can extend deep into the atomic layers [15]. The actions of strains or compositional tuning in TMDCs have been previously investigated. Nevertheless, the interplay of substitutional doping and externally imposed biaxial strain has rarely been studied in terms of pure structural and energetic considerations. [16]. Specifically, the relationship between dopant-induced lattice distortion and strain-dependent formation energetics in ZrS_2 has not been fully understood [17]. A quantitative evaluation of coordination-shell-resolved distortion, strain-modulated stability, and elastic curvature can provide more light on how medium-range structural order is altered by local chemical alteration [18].

In this work, we use a machine-learned interatomic potential framework to analyze the combined effect of biaxial strain and Se substitution in ZrS_2. Our goal is to provide a coherent atomistic picture of dopant-induced lattice relaxation under mechanical disturbance by investigating strain–energy relationships, formation energetics, bond-length redistribution, and shell-matched distortion profiles. The current work offers a mechanically based viewpoint on substitutional

modification in layered dichalcogenides by emphasizing structural stability and elastic responsiveness above electrical characteristics.

2. Computational Methodology

A machine-learned interatomic potential framework built inside the MatterSim [19] environment was used for all atomistic simulations [20]. The used potential allows for effective structural relaxation and energy evaluation for transition metal chalcogenide complexes and was trained using density functional theory (DFT) reference data. In order to reduce false defect interactions under periodic boundary circumstances, the original ZrS2 structure was derived from crystallographic data and enlarged into a 2×2×1 supercell. A quasi-Newton optimization approach was used to perform structural relaxations until the atomic forces were lowered below 10^{-3} eV/Å, unless otherwise noted, all internal atomic coordinates and lattice vectors were let to relax [21].

While keeping periodic boundary conditions, biaxial strain was applied by evenly scaling in-plane lattice parameters within a range of −4% to +4%. Before evaluating energy, complete structural relaxation was carried out for each strain state to guarantee equilibrium atomic configurations [22]. In order to derive curvature coefficients related to in-plane stiffness, the strain–energy relationship was examined using quadratic fitting inside the elastic domain. In order to induce substitutional doping, one sulfur atom was swapped out for a selenium atom in the supercell, which corresponded to a diluted defect concentration [23]. Within the same interatomic potential framework, formation energy was computed in relation to isolated atomic reference energies. By comparing the coordination distances between pristine and doped configurations, shell-matched distortion analysis was carried out, and bond length distributions were calculated using neighbor analysis with a set cutoff radius.

Where appropriate, all reported energies are normalized per atom or displayed per supercell. In order to prevent plastic deformation or structural phase transitions, the applied strain range was selected to stay inside the elastic limit. The formation energy of the substitutional defect was calculated using

$$E_f = E_{tot}^{doped} - E_{tot}^{Pristine} + \mu_S - \mu_{Se}$$

where E_{tot}^{doped} and $E_{tot}^{Pristine}$ represent the total energies of the doped and pristine supercells, respectively. μ_S and μ_{Se} denote the chemical potentials of sulfur and selenium obtained from their isolated atomic reference states.

3. Results and Discussion

3.1 Strain–Energy Response of Pristine and Se-Substituted ZrS_2

Figure 1, shows how the total energy of pristine and Se-substituted ZrS2 varies with the strain applied. In both systems, the trend of the energy-strain relationship is a symmetric parabolic trend in the range of biaxial strain that is −4% to +4%, which implies that the structural response is in the elastic regime. The lowest energy between the two configurations is obtained at zero strain, which makes it clear that the relaxed supercells are equilibrium structures. When comparing the undoped and doped systems a small difference in curvature is noticed. Though the general quadratic behavior can be retained after substitution, the Se-doped structure has a slight alteration in the energy strain sensitivity. This change indicates that the distortion of the local lattice caused by Se introduces subtle changes to the elastic response of the host lattice and does not considerably destabilize the system [17, 24].

While the absolute energy panel (Figure 1B) enables quantitative fitting of the curvature coefficients, the relative energy representation (Figure 1A) emphasizes the same elastic stability window of both systems.

3.2 Energy Landscape Visualization

In order to have a more intuitively understandable view of the structural stability of the system under applied strain, the strain-energy relationship was further modeled as two-dimensional energy landscape as shown in Figure 2. In this illustration, the contour plots of both pure and the Se-substituted ZrS2 systems significantly show a distinct energy basin with the centre of zero strain. The given feature means that the equilibrium position of both structures is at the unstrained position, which is in line with the findings by the energy strain curves. The fact that the energy basins around the origin are almost symmetrical is an indication that the mechanical response of the lattice is balanced under compressive as well as tensile biaxial deformation. Significantly, addition of the Se in the lattice does not seem to cause any major mechanical asymmetry of the system.

Though the overall energy landscape topology of the two systems is almost similar there are slight variations in the steepness and concentration of the energy contours. Specifically, the basin that is related to the Se-substituted structure is characterized by a slightly altered curvature of the system in comparison to the pristine one. This finding is in line with the quadratic fitting analysis which

was illustrated earlier that revealed a slight variation in the elastic response following the substitution. Direct visual comparison of the two contour maps made possible by the use of a common color scale shows that there is a small-scale redistribution of the elastic energy density in the lattice of the doped structure. This can be explained by the fact that the local structural perturbation caused by the Se atoms succeeding a sulfur atom can be explained by the fact that the atomic radius of the Se is bigger than that of sulfur, and this leads to the fact that the surrounding area experiences slight relaxation of the lattice.

Despite this localized distortion, the global symmetry and stability of the host framework remain largely preserved. Overall, these contour maps reinforce the conclusion that Se substitution subtly modifies the elastic characteristics of ZrS2 without compromising its fundamental mechanical stability [25].

Table 1: Comparison of structural parameters.

Material	Zr–S bond length (Å)	Zr–Se bond length (Å)	First shell distortion	Method	Reference
Pristine ZrS2	2.58	–	–	ML potential	This work
Se-doped ZrS2	–	2.71	0.12 Å	ML potential	This work
ZrS2 monolayer	2.55–2.60	–	–	DFT	[7]
Se substituted TMDCs	–	2.68–2.73	0.1 Å	DFT	[26]

3.3 Elastic Curvature Comparison and Strain-Dependent Formation Energetics

The strain-energy data were examined and quantified numerically through quadratic fitting so as to measure the strain response of the system. The in-plane elastic stiffness of the material is represented by the second derivative of the total energy with reference to strain in this method. Figure 3A can be used to compare the fitted energy curves in a narrower strain window, and in this case, subtle differences in the curvature between the pristine and the Se-substituted configurations can be easily seen. Even though both systems behave as expected in regards to the quadratic action in case of elastic deformation, a slight difference in curvature is observed following substitution. The second-derivative profiles (Figure 3B) further illuminate this fact. The profiles indicate that introduction of Se causes a quantifiable, but small alteration in the elastic response of the lattice. Notably, the difference in curvature that is observed is small and does not reflect mechanical softening or structural instability. Rather, it is still in the elastic regime of operation. The findings thus indicate that the substitution is mainly causing local lattice relaxation at the dopant site that

slightly re-distributes the strain energy in the lattice without decreasing the overall structural strength of the material.

By computing the formation energy as a function of biaxial deformation, the stability of Se substitution under external strain was further assessed (Figure 4). As would be expected for dilute defect incorporation, the formation energy stays positive throughout the studied strain window, suggesting that substitution is energetically costly in comparison to isolated atomic references. Nonetheless, there is a modest strain dependence in the formation energy magnitude. Small differences between the tensile and compressive regimes imply that dopant stability may be marginally impacted by external deformation. While no abrupt instability or phase transition is seen within the range under study, this behavior suggests a link between elastic strain and defect energetics. The doped lattice's structural stability under mild mechanical perturbation is confirmed by the comparatively smooth formation energy trend.

3.4 Bond Length Redistribution and Shell-Resolved Distortion Decay

The distribution of bond length of pristine and doped supercells was studied to study the local structural change (Figure 5). The homogeneous coordination within the layered lattice is manifested in the fact that the narrow distribution of undoped Zr-S bonds is concentrated at the center at approximately 2.58 A. As shown by the increased atomic radius of Se, the newly formed Zr-Se bonds shift to longer bond lengths (larger bond lengths of about 2.71 A), when replaced. The small broadening of the surviving Zr-S bonds is a small indication of redistribution of local strain. The statistical comparison shows that replacement does not affect the overall lattice coherence but mostly affects the immediate coordination environment. The presence of a clear Zr-Se peak and no significant fragmentation of the Zr-S distribution indicate a minor local not global distortion.

A shell-matched distortion analysis was carried out to assess the spatial extent of dopant-induced lattice disruption (Figure 6). The distances from the replaced Se atom in the doped configuration and a reference S atom in the pristine lattice were compared. Significant local relaxation around the dopant site is confirmed by the first coordination shell enlargement of about 0.12 Å. Fascinatingly, the fourth shell exhibits a smaller divergence (~0.05 Å), yet similar distortion endures through the second and third shells. This slow decrease suggests that instead of an isolated first-shell action, Se substitution causes a medium-range elastic perturbation.

As higher coordination shells get closer to the undoped reference distances, the distortion stops spreading. According to this discovery, the layered ZrS2 lattice preserves global structural stability by efficiently screening local strain within a constrained radial extent.

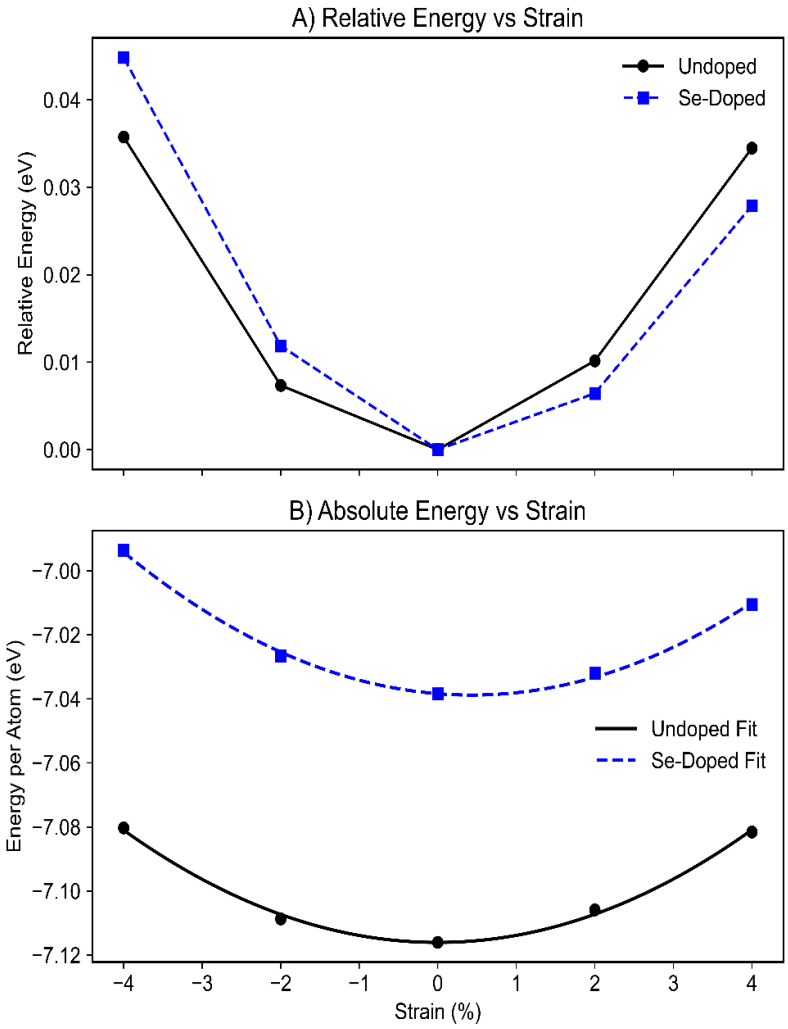

Figure 1: Response of pure and Se-substituted ZrS2 to strain and energy during biaxial deformation. (A) Variation in relative energy with regard to equilibrium configuration. (B) Elastic curvature is extracted using absolute energy per atom and quadratic fitting.

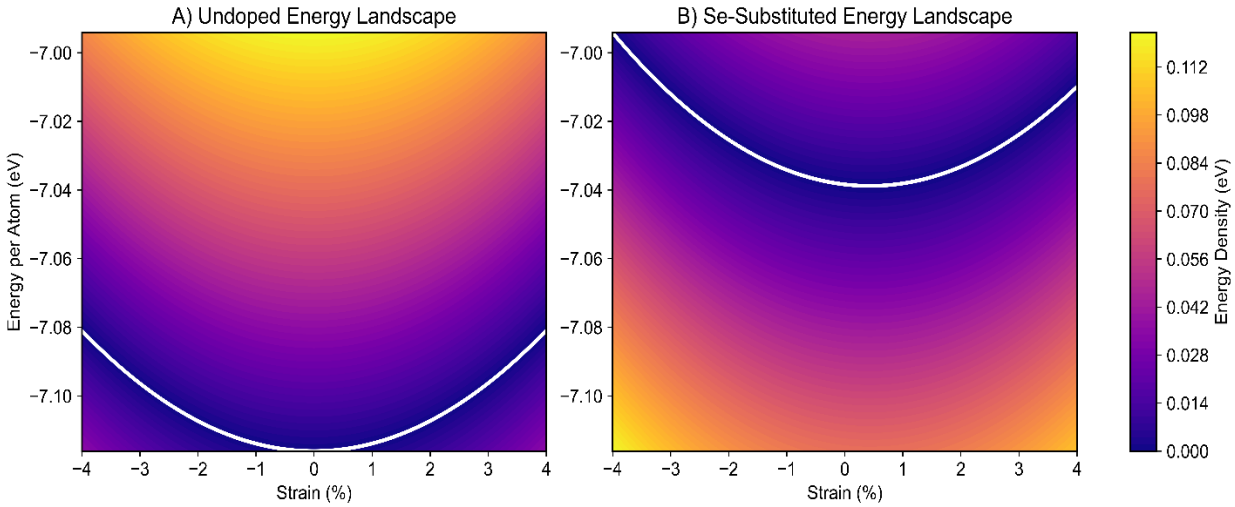

Figure 2: Pristine and Se-substituted ZrS2 two-dimensional energy landscape under biaxial strain. The strain-dependent energy basin is depicted in contour plots, which highlight variations in curvature and elastic response.

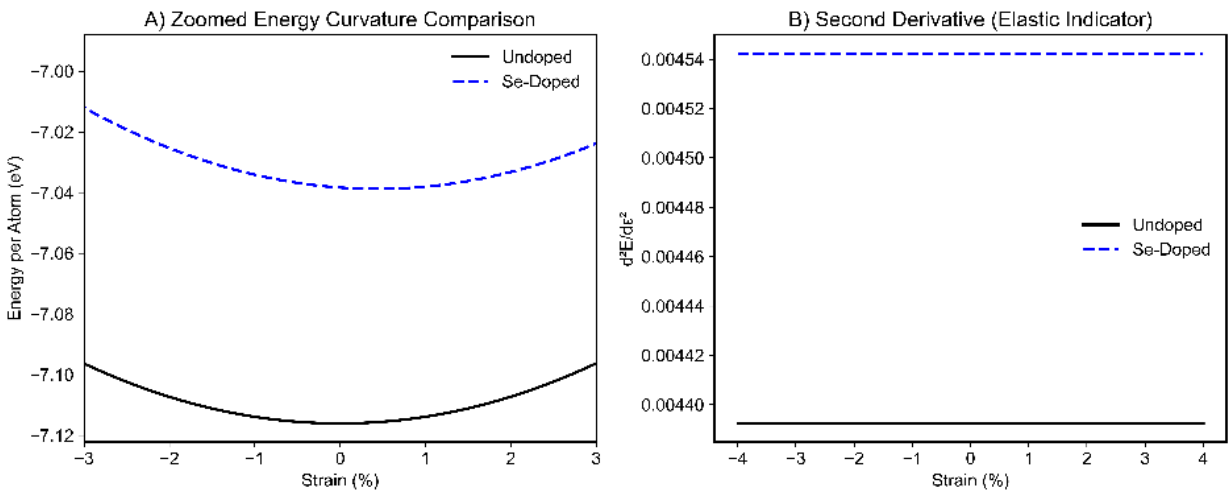

Figure 3: Quadratic fitting of strain-energy data yields a comparison of elastic curvatures. (A) Fitted energy curve comparison in zoomed-in. (B) Second derivative profile showing the difference in relative stiffness between doped and virgin structures.

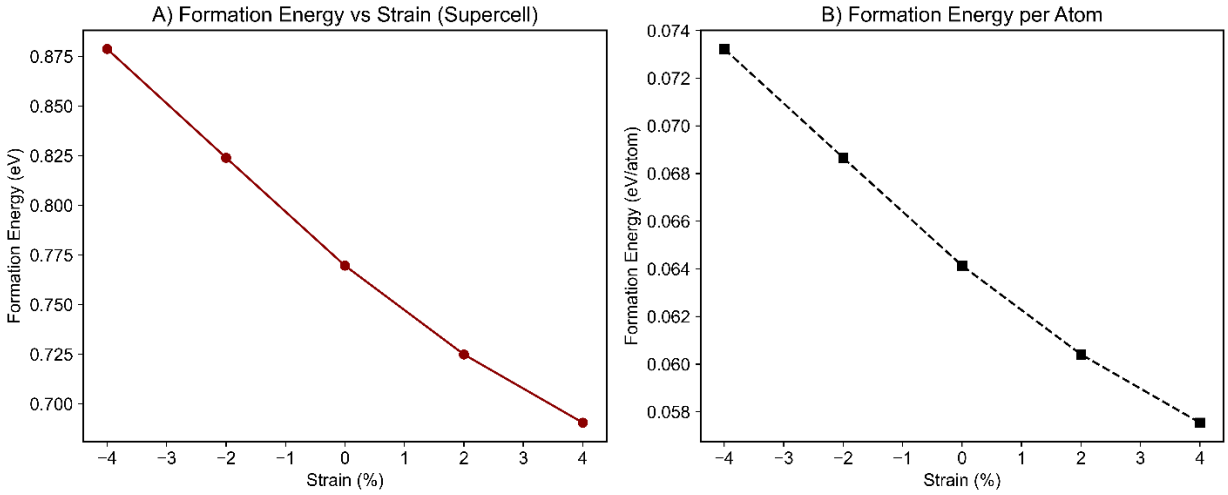

Figure 4: Se substitution formation energy in ZrS_2 as a function of biaxial strain. (A) Total energy of creation for each supercell. (B) Normalized formation energy per atom, showing the energetic connection between strain and defects.

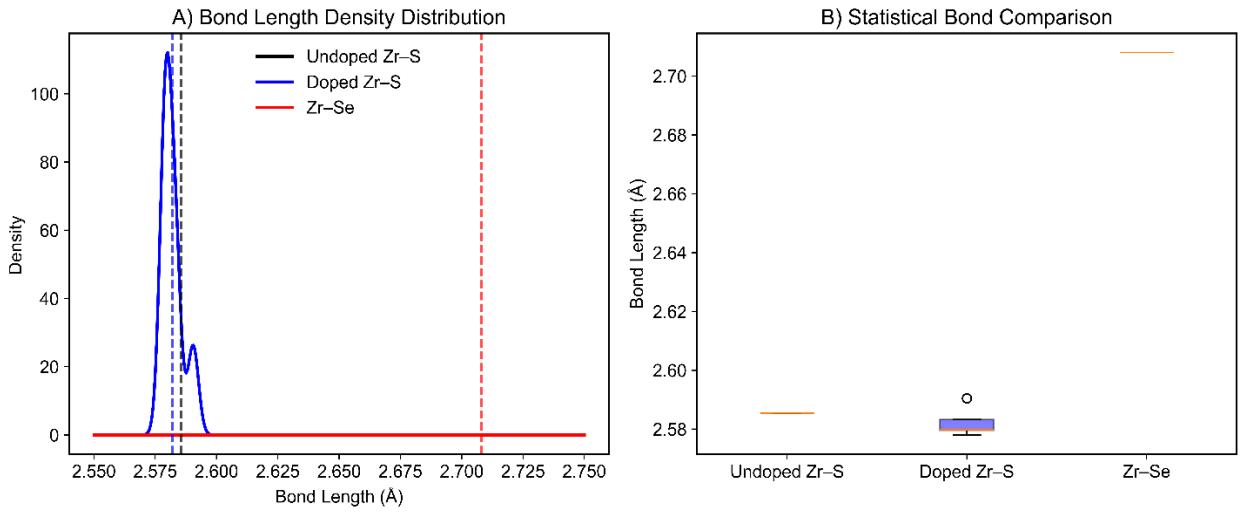

Figure 5: Analysis of the bond length distribution. (A) Zr–S and Zr–Se bond density distribution. (B) A statistical comparison of bond lengths demonstrates how Se substitution causes local expansion.

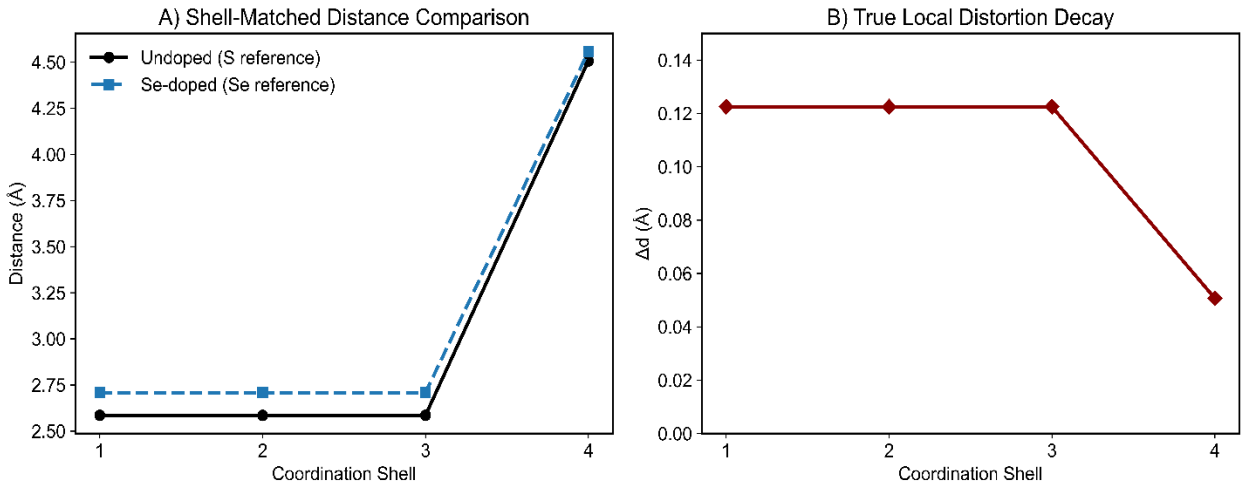

Figure 6: The distortion decay profile surrounding the Se dopant is shell-matched. (A) Coordination-shell distances between doped and virgin structures are compared. (B) Medium-range elastic relaxation is indicated by the radial decline of lattice distortion.

Conclusion

This work uses a machine-learned interatomic potential framework to systematically investigate the structural response of ZrS2 to substitutional Se doping under biaxial strain. The applied strain range stays within the elastic region, as confirmed by the strain–energy relationship of both pristine and doped systems, which shows a well-defined quadratic dependence within the studied ±4% deformation window. Se substitution modifies ZrS2 elastic curvature in a quantifiable but mild way. Quantitative fitting shows a redistribution of strain sensitivity in the doped arrangement, but the lattice overall mechanical stability is maintained. Additionally, strain-dependent formation energy analysis shows that external deformation has a negligible impact on dopant stability, indicating a link between lattice strain and defect energetics that does not result in structural instability. In accordance with atomic size differences, bond length distribution analysis shows a distinct expansion of Zr–Se bonds in comparison to Zr–S bonds. The dopant-induced lattice expansion is not strictly limited to the first coordination shell, but rather extends to the fourth shell with progressively diminishing magnitude, according to shell-matched distortion analysis. Instead of a purely localized defect disruption, this behavior suggests a medium-range elastic relaxation mechanism.

Altogether, the combined structural study, which includes mechanistic, thermal, and shell-resolved methods, has a cohering description of substitutional change in layered ZrS2. In the findings, it is

proved that under mild biaxial strain, Se incorporation leads to the regulated lattice relaxation and structural integrity is maintained. In this work, it is demonstrated that machine-learning interatomic potentials can be effectively utilized to investigate strain-defect relations in layered transition metal dichalcogenides.

Reference

1. K.S. Novoselov, et al., Electric field effect in atomically thin carbon films. science, **306**(5696) (2004).666-669.
2. M. Chhowalla, et al., The chemistry of two-dimensional layered transition metal dichalcogenide nanosheets. Nature chemistry, **5**(4) (2013).263-275.
3. Q.H. Wang, et al., Electronics and optoelectronics of two-dimensional transition metal dichalcogenides. Nature nanotechnology, **7**(11) (2012).699-712.
4. J.A. Wilson and A. Yoffe, The transition metal dichalcogenides discussion and interpretation of the observed optical, electrical and structural properties. Advances in Physics, **18**(73) (1969).193-335.

5. R. Wu, et al., Synthesis, modulation, and application of two-dimensional TMD heterostructures. Chemical Reviews, **124**(17) (2024).10112-10191.
6. V. Brune, et al., Molecular Precursors to Group IV Dichalcogenides MS2 (M= Ti, Zr, Hf). Zeitschrift für anorganische und allgemeine Chemie, **648**(23) (2022).e202200049.
7. F.A. Rasmussen and K.S. Thygesen, Computational 2D materials database: electronic structure of transition-metal dichalcogenides and oxides. The Journal of Physical Chemistry C, **119**(23) (2015).13169-13183.
8. J.O. Island, et al., Environmental instability of few-layer black phosphorus. 2D Materials, **2**(1) (2015).011002.
9. H. Zhan, D. Guo, and G. Xie, Two-dimensional layered materials: from mechanical and coupling properties towards applications in electronics. Nanoscale, **11**(28) (2019).13181-13212.

10. K. He, et al., Experimental demonstration of continuous electronic structure tuning via strain in atomically thin MoS2. Nano letters, **13**(6) (2013).2931-2936.
11. H.J. Conley, et al., Bandgap engineering of strained monolayer and bilayer MoS2. Nano letters, **13**(8) (2013).3626-3630.
12. H.P. Komsa and A.V. Krasheninnikov, Engineering the electronic properties of two-dimensional transition metal dichalcogenides by introducing mirror twin boundaries. Advanced Electronic Materials, **3**(6) (2017).1600468.
13. W. Zhou, et al., Intrinsic structural defects in monolayer molybdenum disulfide. Nano letters, **13**(6) (2013).2615-2622.
14. J. Hong, et al., Exploring atomic defects in molybdenum disulphide monolayers. Nature communications, **6**(1) (2015).6293.
15. H. Li, et al., From bulk to monolayer MoS2: evolution of Raman scattering. Advanced Functional Materials, **22**(7) (2012).1385-1390.

16. S. Chandrasekaran, et al., Publications Repository-Helmholtz-Zentrum Dresden-Rossendorf. Computing in Science & Engineering, **20**(4) (2018).95-106.
17. P. Johari and V.B. Shenoy, Tuning the electronic properties of semiconducting transition metal dichalcogenides by applying mechanical strains. ACS nano, **6**(6) (2012).5449-5456.
18. J. Yang, et al., Anisotropic heat transfer properties of two-dimensional materials. Nanotechnology, **32**(16) (2021).162001.
19. H. Yang, et al., Mattersim: A deep learning atomistic model across elements, temperatures and pressures. arXiv preprint arXiv:2405.04967, (2024).
20. J. Behler, Perspective: Machine learning potentials for atomistic simulations. The Journal of chemical physics, **145**(17) (2016).
21. R. Drautz, Atomic cluster expansion for accurate and transferable interatomic potentials. Physical Review B, **99**(1) (2019).014104.

22. L.D. Landau, et al., Theory of elasticity: volume 7. Vol. 7. 2012: Elsevier.
23. C. Freysoldt, et al., First-principles calculations for point defects in solids. Reviews of modern physics, **86**(1) (2014).253-305.
24. Z. He, et al., Atomic structure and dynamics of metal dopant pairs in graphene. Nano letters, **14**(7) (2014).3766-3772.
25. H.-P. Komsa and A.V. Krasheninnikov, Native defects in bulk and monolayer MoS 2 from first principles. Physical Review B, **91**(12) (2015).125304.
26. L. Nguyen, et al., Atomic defects and doping of monolayer NbSe2. ACS nano, **11**(3) (2017).2894-2904.